ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

MID-SEM EXAMINATION — DECEMBER / JUNE 2024

FIRST YEAR (COMMON TO ALL BRANCHES) — SEMESTER 1

Time Allowed: 1.5 Hours **Maximum Marks: 30**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- PART A ---

Q.1 State Kirchhoff's Laws (KCL, KVL). State and prove Thevenin's and Norton's Theorems. [15 Marks]

[OR]

Q.1 (Alt) State Maximum Power Transfer Theorem for DC circuits. Solve bridge circuit using Mesh analysis. [15 Marks]

--- PART B ---

Q.2 Define RMS, Average, Form factor of AC sine wave. Explain single-phase transformer EMF equation. [15 Marks]

[OR]

Q.2 (Alt) Calculate primary turns, flux, and full-load currents for a 10 kVA 2200/220V transformer. [15 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****